



VAYURON ADVANCED SYSTEMS

Advanced Unmanned & Tactical Aerial Systems



Ghost Stalker — Long-Range VTOL ISR Platform

Vayuron Advanced Systems

Ghost Stalker -Long-Range VTOL ISR Platform

Persistent • Autonomous • Runway-Independent



Fixed-Wing VTOL | 2.9 m Wingspan | Up to 90 min Endurance | Up to 50 km Radius | 5 KG Payload

PRODUCT PROFILE & TECHNICAL SPECIFICATION SHEET

Made in India | OEM Manufacturer | Defence Supplier

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1. PRODUCT OVERVIEW

Ghost Stalker is a fixed-wing VTOL (Vertical Take-Off and Landing) Intelligence, Surveillance & Reconnaissance (ISR) platform engineered for long-endurance monitoring, target observation, and autonomous aerial missions. The platform combines the operational flexibility of vertical take-off and landing with the aerodynamic efficiency of fixed-wing flight, enabling deployment from confined environments without requiring runways, catapults, or recovery infrastructure.

Designed for defence, homeland security, industrial monitoring, border surveillance, and critical-infrastructure protection, Ghost Stalker delivers persistent situational awareness across extended operational areas. With an endurance of up to **90 minutes**, an operational radius of up to **50 km**, and a **payload capacity of up to 5 kg**, the platform is capable of carrying a wide range of ISR payloads, including EO/IR cameras, thermal imaging systems, AI-enabled target detection modules, communication relays, and specialized mission equipment.



Figure 1 — Ghost Stalker long-range VTOL ISR platform, 2.9 m carbon-fibre composite airframe.



2. OPERATIONAL ADVANTAGES

- Runway-independent deployment via vertical take-off and landing
- Long-duration surveillance missions, up to 90 minutes endurance
- Rapid launch and recovery from confined environments
- Autonomous waypoint navigation
- Low logistical footprint
- Portable ground-control architecture
- Real-time intelligence collection

3. TECHNICAL SPECIFICATIONS

All figures are nominal and subject to configuration, payload and environmental conditions. Final performance is confirmed against the agreed end-user requirement at point of supply.

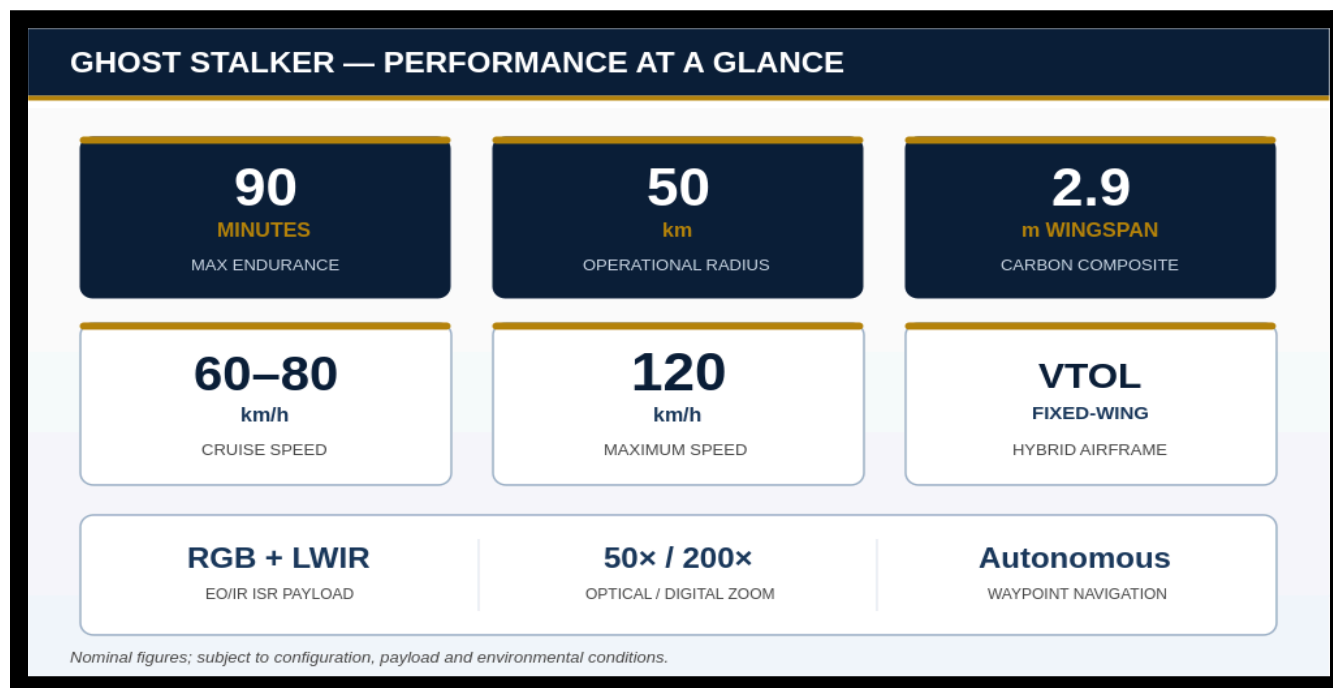


Figure 2 — Ghost Stalker performance at a glance.



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AIRFRAME & ENDURANCE

Aircraft Type	Fixed-Wing VTOL UAV
Mission Class	ISR / Surveillance
Airframe Material	Carbon Fibre Composite
Wingspan	2.9 m
Endurance	Up to 90 minutes
Operational Radius	Up to 50 km
Payload	5 KG

PERFORMANCE & FLIGHT SYSTEMS

Cruise Speed	60 – 80 km/h
Maximum Speed	120 km/h
Take-Off Method	Vertical
Landing Method	Vertical
Navigation	GNSS-Assisted Autopilot
Mission Planning	Autonomous Waypoint Based
Communication Link	Encrypted Digital Telemetry
Payload Capacity	5 KG

4. VTOL FLIGHT PROFILE

Ghost Stalker launches vertically, transitions to efficient fixed-wing flight for transit and on-station loiter, then recovers vertically into confined areas — combining the infrastructure independence of a rotorcraft with the endurance and range of a fixed-wing aircraft.

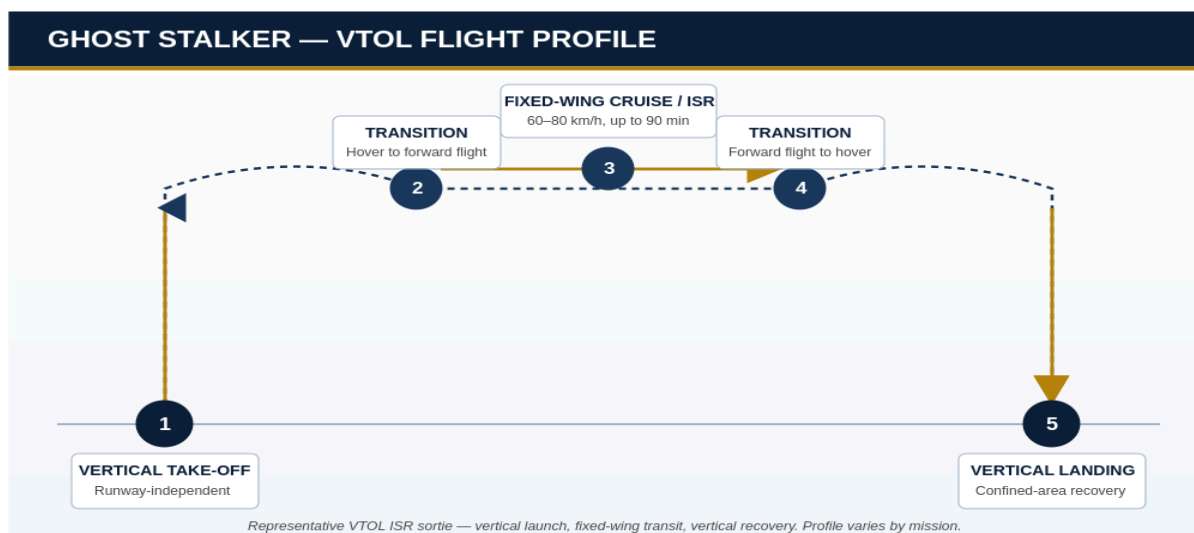


Figure 3 — Representative VTOL ISR sortie, vertical launch through vertical recovery.

5. SYSTEM ARCHITECTURE

The Ghost Stalker system comprises an air-vehicle segment and a portable ground-control segment, linked by an encrypted digital telemetry datalink for command, control, telemetry, and live video.

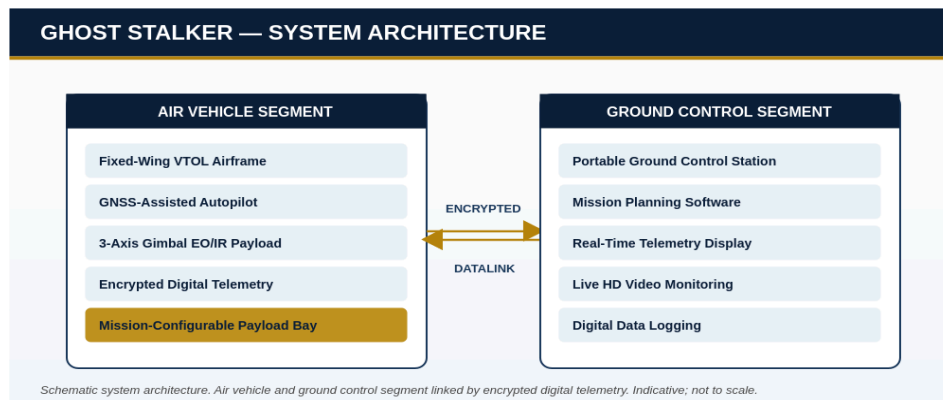


Figure 4 — Ghost Stalker system architecture: air vehicle and ground control segment.

6. ISR SENSOR PAYLOAD

Ghost Stalker carries a stabilised electro-optical / infrared (EO/IR) ISR payload for day-and-night surveillance and target observation:

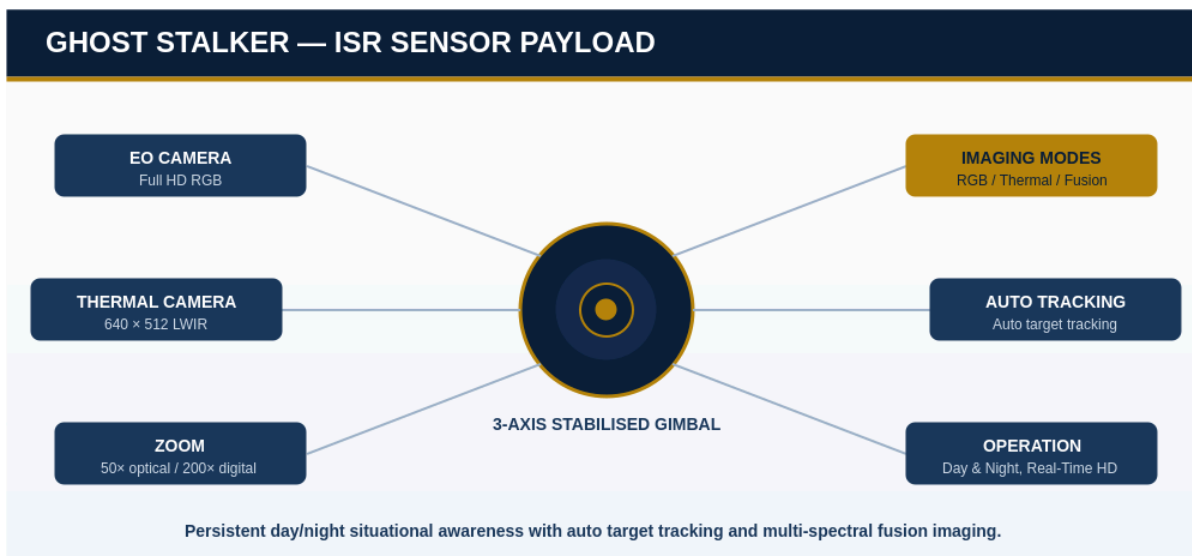


Figure 5 — Ghost Stalker stabilised EO/IR ISR sensor capability.

ISR PAYLOAD SPECIFICATIONS	
EO Camera	Full HD RGB
Thermal Camera	640 × 512 LWIR
Optical Zoom	50×
Digital Zoom	200×
Imaging Modes	RGB / Thermal / Fusion
Tracking	Auto Target Tracking
Stabilisation	3-Axis Gimbal
Video Output	Real-Time HD
Operation	Day & Night
Mission Role	ISR & Surveillance



7. GROUND CONTROL SEGMENT

The platform is operated through a portable ground-control segment comprising:

- Portable ground control station
- Mission planning software
- Real-time telemetry display
- Video monitoring interface
- Digital data logging

8. APPLICATIONS

Border Surveillance - Persistent monitoring of international borders, coastal regions, and sensitive areas to detect unauthorized movement, infiltration attempts, smuggling activities, and potential security threats. Ghost Stalker provides real-time intelligence and continuous situational awareness across large operational sectors.

Tactical Reconnaissance - Supports military, paramilitary, and law-enforcement operations with real-time aerial intelligence, route reconnaissance, target observation, battlefield awareness, and force-protection missions. The platform can be deployed rapidly to gather actionable intelligence before and during operations.

Intelligence, Surveillance & Reconnaissance (ISR) - Provides persistent ISR capability for defence, homeland security, strategic installations, and critical missions requiring long-endurance observation, target tracking, and real-time intelligence collection.

Communications Relay Missions - Can be configured as an airborne communications relay platform to extend communication coverage in remote, mountainous, disaster-affected, or communication-denied environments.

Search and Rescue (SAR) - Assists emergency teams in locating missing persons, stranded individuals, and disaster victims across large geographical areas. Thermal and EO payloads enable operations during both day and night conditions.

Maritime & Coastal Surveillance - Suitable for monitoring coastal regions, harbors, waterways, and maritime infrastructure. The platform can assist in detecting unauthorized vessels, monitoring fishing activities, and supporting coastal security operations.



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ABOUT THE COMPANY

VAYURON ADVANCED SYSTEMS PRIVATE LIMITED

Engineering Intelligent Systems for Mission-Critical Operations.

Vayuron Advanced Systems Private Limited is an indigenous developer and OEM manufacturer of advanced unmanned and intelligent systems, purpose-built for defence, homeland security, utilities, and critical infrastructure. The company designs, develops, and manufactures mission-critical technologies that deliver decisive operational advantage in demanding and contested environments.

Vayuron's work spans the full spectrum of intelligent aerial and autonomous capability, with core competencies in ISR, tactical UAVs, FPV systems, autonomous systems, AI-enabled vision, and secure communications, extending to survey and mapping, utility inspection, and asset intelligence. By integrating these disciplines on indigenous platforms, Vayuron equips frontline forces, security agencies, and infrastructure operators with reliable, sovereign, and rapidly deployable solutions.

Committed to self-reliance in defence and strategic technologies, Vayuron Advanced Systems combines rigorous engineering, modular architectures, and field-proven design philosophy to build systems that perform when it matters most.

Email: info@vayuronadvancedsystems.com

Bhopal, Madhya Pradesh, India

DISCLAIMER & TERMS OF REFERENCE

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Customisation and Final Configuration: Customisation of the platform, payload integration, avionics, communications, software, and all other product features for the actual delivered product will be undertaken as per mutual agreement with the procurer. Final configuration, technical parameters, acceptance criteria, scope of supply, and applicable standards will be defined and confirmed contractually against the agreed end-user requirement.

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